

A New Clue to the Puzzle of the Expanding Universe

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Scientists know the universe is expanding, but they disagree about how fast. Two of the best ways to measure the speed, called the **Hubble constant**, give answers that do not match. This puzzle is known as the Hubble tension.

One method uses the cosmic microwave background, the faint afterglow of the Big Bang. Telescopes such as the **Planck Space Telescope** measured tiny patterns in this ancient light. Using a model of the universe, those patterns predict an expansion rate of about 67 kilometers per second per megaparsec.

The second method is more direct. It measures how fast faraway galaxies move away from us, using the brightness of exploding stars called **supernovae**. Because these **supernovae** have a known brightness, how dim they look shows how far away they are. Observations with the Hubble and James Webb space telescopes give a higher value of about 73. The difference may seem small, but it is statistically very significant. If both methods are correct, the standard model of the universe may be missing something.

A team of researchers wondered whether tiny magnetic fields from the very early universe could help. Scientists have studied these "**primordial**" fields for years. The team's idea is that such fields would have changed how the first hydrogen formed, making matter slightly clumpy. That would shift the moment the universe became transparent. It would also change the "**cosmic ruler**" used to measure distances, which could ease the tension.

For the first time, the team ran full three-dimensional **simulations** of the early universe with magnetic fields and tested the results against observations. Their idea remained possible. They found a mild preference for these fields, which they say is not yet a discovery, but a meaningful hint.

Word Treasure

Hubble constant -- The number that describes how fast the universe is expanding.

Planck Space Telescope -- A telescope that measured tiny patterns in the faint afterglow of the Big Bang.

supernovae -- Exploding stars.

primordial -- From the very earliest times.

simulations -- Computer programs that copy how something works.

cosmic ruler -- The pattern used to measure distances in the universe.

Break it down

Opening: scientists disagree about the expansion speed

Method one: the cosmic microwave background

Method two: supernovae and space telescopes

The Hubble tension

Primordial magnetic fields and the new idea

Simulations and a mild, early hint

Why it matters

If both measurements are right, the standard model of the universe must be missing something important, so a new idea like this could lead to better understanding.

Test what you remember

1. What is the Hubble constant used for?

- ☐ (A) Measuring the size of black holes
- ☐ (B) Measuring how fast the universe is expanding
- ☐ (C) Counting the planets in the Solar System
- ☐ (D) Timing the Moon's orbit

2. What expansion rate do patterns in the cosmic microwave background predict?

- ☐ (A) About 31
- ☐ (B) About 67
- ☐ (C) About 73
- ☐ (D) About 120

3. Why do supernovae help with the second method?

- ☐ (A) They are the biggest stars
- ☐ (B) They make galaxies move
- ☐ (C) They give off magnetic fields
- ☐ (D) Their known brightness shows how far away they are

4. What do the researchers say the difference between the two values is?

- ☐ (A) Statistically very significant
- ☐ (B) Too small to matter
- ☐ (C) An error in the telescopes
- ☐ (D) Caused by the Moon

5. What did primordial magnetic fields possibly change?

- ☐ (A) How the first hydrogen formed in the early universe
- ☐ (B) The shape of Earth
- ☐ (C) The brightness of the Sun
- ☐ (D) The path of the Planck telescope

6. What was new about the team's simulations?

- ☐ (A) They only looked at the Moon
- ☐ (B) They were done by telescopes in space
- ☐ (C) They were full three-dimensional runs of the early universe with magnetic fields
- ☐ (D) They used no computers

Answer key (try the quiz first!): 1: B 2: B 3: D 4: A 5: A 6: C

Questions to think about

1. The research team: Say the data show a mild preference for primordial magnetic fields, a meaningful hint rather than a discovery.

MY THOUGHTS (at least 20 words):